Project Dataset: Windows Malware

Windows Malware Detection Research Project

Group No in Track Sheet: 4

Group Members

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Focus Area: Static PE-header features • 1D-CNN • Attention-based Malware Detection

Research Track 3: CNN+Attention-based model design for Windows malware classification

Task 1: Related Work Review and Research Gap

We are working with the SOMLAP dataset from Kaggle (ravikiranvarmap/somlap-data-set). It contains 51,409 Windows PE files: 19,809 malware samples collected from VirusShare and 31,600 benign executables and DLLs taken from a Windows 10 installation. Every file is described by 108 attributes extracted purely from the PE headers (DOS header, File header and Optional header fields), so the data is tabular and fully static; nothing has to be executed to get the features.

The dataset comes from Kattamuri et al. (2023). In that paper the authors reduced the 108 attributes with swarm algorithms and then ran classical classifiers on top, and their best result was 99.37% accuracy using just 12 features picked by ant colony optimization. A very recent follow-up (Alwateer et al., 2026) pushed this to 99.89% with a hierarchical deep network, still without attention. These two numbers are the bars our model has to clear.

Since our group is in Track 3, our plan is a 1D-CNN with an attention block, trained directly on all 108 header features. The next section reviews ten papers from 2022 to 2026. We deliberately mixed three kinds of work: papers on SOMLAP itself, papers on other tabular PE or memory features, and the image-based CNN + attention papers where most attention research in this area has happened. The last column gives the attention type, as required for our track.

Related-work table

Title	Authors	Year	Dataset	Application area	Method / model	Metrics (reported)	Strengths	Limitations	Research gap	Relation to our work	Attention type (Track 3)
MSAAM: A Multiscale Adaptive Attention Module for IoT Malware Detection and Family Classification	Wang C., Zhao Z., Wang F., Li Q.	2022	Malimg + Microsoft BIG 2015 (grayscale images)	IoT malware detection and family classification	CliqueNet with the MSAAM module swapped in for its usual attention block	Detection 99.8%; families 99.2% (Malimg) and 98.2% (BIG 2015)	The module tunes its own mix of channel vs. spatial attention; copes well with imbalanced families and obfuscation	Heavy architecture; detection tested on a fairly small subset (1,087 malware); images only	Attention module never tried outside image benchmarks	The channel/spatial weighting idea carries over to weighting groups of PE-header fields	Multiscale adaptive channel + spatial attention
Swarm Optimization and Machine Learning Applied to PE Malware Detection towards Cyber Threat Intelligence	Kattamuri S.J., Penmatsa R.K.V., Chakravarty S., Madabathula V.S.P.	2023	SOMLAP (51,409 files, 108 PE-header attributes; VirusShare + Windows 10 benign)	Windows PE malware detection (static, tabular)	Feature selection with ACO, CSO and GWO, then classical ML (DT, RF, SVM, kNN, NB)	Best 99.37% accuracy, ACO with only 12 features	This is the paper behind our dataset; careful feature-reduction study; clean header-only attributes	Classical ML only, so no deep learning at all; drops 96 of 108 features before classifying	No deep or attention-based model tested on SOMLAP	Our dataset and our first baseline: 99.37% is the number to beat	None
Attention-based convolutional neural network deep learning approach for robust malware classification	Ravi V., Alazab M.	2023	Malware images, 25 families (Malimg-style)	Malware family classification from images	CNN with multi-headed attention that highlights infected image regions	About 99% accuracy over 25 families, ahead of the plain-CNN versions	One of the cleaner demonstrations that adding attention to a CNN actually pays off	Byte-to-image conversion throws away header semantics; benchmark close to saturated	Attention only on images, not on structured features	Backs our main hypothesis: attention on top of a CNN improves malware classification	Multi-head self-attention fused with CNN features
A Malicious Code Detection Method Based on Stacked Depthwise Separable Convolutions and Attention Mechanism	Huang H., Du R., Wang Z., et al.	2023	Malimg + an augmented "Blended+" image set	Malware detection and classification from images	CoAtNet: stacked depthwise-separable convolutions combined with self-attention	99.33% (Malimg), 96.60% (Blended+); beats VGG16, ResNet50, EfficientNetB0 and others	Lightweight convolutions plus attention give a good capacity/generalization trade-off	Accuracy drops noticeably on the harder blended set; image pipeline only	Conv + attention hybrids untested on feature-vector data	Closest architectural template to ours; our 1D-CNN + attention is basically its tabular cousin	Self-attention (transformer block) mixed with convolutions
MAD-ANET: Malware Detection Using Attention-Based Deep Neural Networks	Al-Ghanem W.K., Qazi E.U.H., Zia T., et al.	2025	CIC-MalMem-2022 (memory-dump features, tabular)	Obfuscated malware detection from memory analysis	Attention-based DNN-CNN, binary and multi-class	99.5% binary; 97.9% multi-class; precision 97.42%, recall 97.95%	Rare example of attention applied to tabular security features, so the closest setting to ours	Memory features need the sample to be executed first; multiclass clearly weaker; one dataset	Attention on static PE-header features still untried	Evidence that CNN + attention works on tabular cyber data at all	Feature-attention layer inside a DNN-CNN
GCSA-ResNet: a deep neural network architecture for Malware detection	Fan Y., Zhang K., Zheng B., et al.	2025	Malimg + Microsoft BIG 2015 (grayscale images)	Malware detection and family classification from images	ResNet with Global Channel-Spatial Attention (GCSA) blocks	Around 99.1% (Malimg) and 98.6% (BIG 2015); cuts false positives by 40-50% vs. baselines	Actually reports false-positive rate, which most of this literature skips	Image-only again; both benchmarks are nearly maxed out; nothing on unseen samples	Never evaluated on non-image malware data	Its channel-attention design is a candidate for our attention block over 108 header fields	Global channel + spatial attention (SE-style) on ResNet
Binary and multiclass malware classification of windows portable executable using classic machine learning and deep learning	Belgacem M.B.	2025	Three public Kaggle PE datasets (header features + one image set)	Windows PE malware, binary and multiclass	Head-to-head comparison of four classical ML and four deep models	Random forest and ANN come out best at roughly 99%; SVM and naive Bayes worst	Practical guidance on which model suits a given dataset size and feature count	Only plain architectures; results swing a lot from dataset to dataset	The comparison stops before attention-based models	Tells us plain DL does not automatically beat ML on header data, which is exactly why we add attention	None
EMBER2024 - A Benchmark Dataset for Holistic Evaluation of Malware Classifiers	Joyce R.J., Miller G., Roth P., et al.	2025	EMBER2024: 3.2M files, six formats, plus an "evasive" challenge set	Benchmarking of static malware classifiers	New benchmark dataset + baseline classifiers (EMBER feature v3)	Benchmark paper; baselines degrade sharply on the AV-evasive challenge set	Huge scale, seven tasks, and the first challenge set built from initially-undetected malware	Not a model contribution; features go beyond pure PE headers	Static classifiers that look great on easy benchmarks struggle on evasive samples	A caution for us: high accuracy on SOMLAP must be discussed alongside generalization	None

Title	Authors	Year	Dataset	Application area	Method / model	Metrics (reported)	Strengths	Limitations	Research gap	Relation to our work	Attention type (Track 3)
Research on image malware classification method based on AlexNet transfer learning and MHSA attention mechanism	Hou X.	2026	MaleVis (malware images)	Malware family classification from images	Pre-trained AlexNet, frozen lower layers, MHSA added at the 5th conv layer; Grad-CAM for inspection	87.3% accuracy on MaleVis	Very recent; shows MHSA lifts a transfer-learned CNN and uses Grad-CAM to show where the model looks	Accuracy is modest; short conference paper; images only	Transfer + attention still weak on harder image sets	Fresh (2026) confirmation that bolting attention onto a CNN helps, and a reminder harder data hurts	Multi-head self-attention (MHSA)
A hierarchical deep learning framework with doubly regularized loss for robust malware detection and family categorization	Alsaedi S.A., Alwateer M., Alharbi N.H., et al.	2026	BODMAS, SOMLAP and CLaMP (tabular PE features)	Windows PE detection, then family categorization	Two-stage hierarchical deep network trained with a doubly regularized loss	Weighted accuracy up to 99.89% (binary) and 96.62% (families)	One of the only DL papers tested on SOMLAP itself; three datasets; loss design is well argued	No attention, and no view of which header fields drive a prediction; family stage weaker	Deep nets beat classical ML on SOMLAP, but stay a black box	The current best deep result on our exact dataset; 99.89% is the real bar for our model	None

Research gap

Reading the ten papers side by side, a clear pattern shows up. When researchers want attention, they turn the binary into a grayscale image and use image attention. When they work on header features like ours, they stick to classical ML or plain deep networks. The two lines of work barely touch, and that intersection is where our project sits.

Gap 1. Attention has not been applied to PE-header features.

Wang et al. (2022), Ravi and Alazab (2023), Huang et al. (2023), Fan et al. (2025) and Hou (2026) all apply attention to malware images. The image conversion step is exactly the problem: once a binary becomes pixels, fields like SizeOfCode or DllCharacteristics lose their meaning, and an attention heatmap over pixels tells an analyst very little. Nobody in our review runs a CNN with attention directly on a structured header-feature vector.

Gap 2. SOMLAP itself has no attention-based result.

The dataset paper only used classical ML (99.37% with ACO picking 12 of 108 features). Alwateer et al. (2026) showed a deep network can reach 99.89%, which proves there is headroom beyond classical ML, but their model has no attention either. So whether attention can match that accuracy while keeping all 108 features in play is an open and very testable question.

Gap 3. No previous study explains which header fields matter per prediction.

Feature selection in the baseline says which 12 features survived, but that is a global, one-off answer. The deep model of Alwateer et al. gives no answer at all. Attention weights over named header attributes would give a per-sample explanation an analyst can actually read, which both SOMLAP papers list as a motivation (threat intelligence) but neither delivers.

Gap 4. The image benchmarks are saturated and hide the metrics that matter.

Malimg and BIG 2015 results now sit above 98-99% across the board (99.8% for MSAAM, 99.33% for CoAtNet, about 99.1% for GCSA-ResNet), so differences between architectures are basically noise. On harder data the picture changes fast: Hou (2026) gets only 87.3% on MaleVis, and EMBER2024 shows baselines collapsing on AV-evasive samples. False-positive rate, which decides whether a detector is usable in practice, is reported almost nowhere; GCSA-ResNet is the exception.

Gap 5. Attention on tabular security data is still in its infancy.

MAD-ANET (Ahmad, 2025) is the one paper we found that applies attention to tabular features, and it needed memory dumps, meaning each sample must be executed in a sandbox first. Header features are free by comparison: they come from a static parse of the file. That cheaper, safer setting is exactly where attention has not been tried.

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